AgriPower PH: Turning Harvest Waste into Clean Energy

Mini Project Documentation Report

In Partial Fulfillment of the Requirements for:

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Introduction

The Philippines produces millions of tons of agricultural residues annually from rice, coconut, and sugar industries. Much of this material decomposes in fields or undergoes open burning rather than conversion into electricity for homes and industries. Biomass currently accounts for approximately 13% of the nation's energy consumption. This establishes it as the predominant renewable energy source after geothermal power. Substantial untapped potential persists due to inefficiencies in collection and market coordination (EnergyTracker Asia, 2024). This issue impacts rural farmers who forfeit potential revenue streams. Local government units grapple with waste-related pollution. The country confronts energy security vulnerabilities amid escalating fossil fuel imports (DOE Philippines, 2024). Primarily affected are smallholder farmers, operators of biomass power facilities, and municipal planning authorities.

Problem Description

Agricultural byproducts such as rice husks (approximately 4 million tons annually), coconut shells (3.5 million tons), and sugar bagasse (5 million tons) possess the capacity to generate 150 megawatts of electricity. This assumes systematic collection and supply to the nation's more than 20 operational biomass power plants (ASEAN Briefing, 2021). Presently the majority of these residues undergo open combustion. This emits methane and particulate matter. Residues accumulate in landfills and impose environmental costs while forgoing economic benefits. The Department of Energy reports that approved biomass capacity stands at 182 megawatts. Operational utilization remains suboptimal owing to inconsistent feedstock availability (EnergyTracker Asia, 2024).

The Philippines leaves 16 million tons of residues unexploited each year (World Bioenergy Association, 2024). Regions such as Negros Occidental and Regions IV-A and VI concentrate 60% of this latent capacity. Coordination deficiencies persist (DOE Philippines, 2024). Farmers derive no income from residues. Power plants incur elevated costs for sporadic supplies. Compounding these issues are inherent limitations. These include the fragmented distribution of residues across thousands of

barangays managed by smallholder farmers who constitute 80% of producers. Prohibitive logistics expenses consume 40 to 60% of potential profits for low-density materials transported over 20 to 50 kilometers. Informational asymmetries exist wherein plants lack visibility into residue availability and farmers remain unaware of prospective buyers or prevailing market rates.

Proposed Solution

AgriPower PH constitutes a web- and mobile-based platform. It leverages geographic information systems (GIS) and algorithmic matching to integrate agricultural residue suppliers, namely farmers and milling operations, with biomass power plants and cooperatives. This rectifies deficiencies across the supply chain.

The platform incorporates four principal features. A GIS residue mapping interface furnishes real-time heatmaps. These delineate husk and bagasse volumes by barangay. It integrates data from the Department of Agriculture alongside geolocated farmer submissions. A smart matching mechanism enables power plants to specify requirements, for example "200 tons of bagasse at ₱2 per kilogram." An algorithm that prioritizes suppliers based on proximity and volume availability. A logistics calculator processes inputs such as truck capacity and fuel costs. It yields projections like "20-kilometer haul at ₱800 per ton yielding ₱1,400 profit." Lastly, an analytics dashboard affords local government units visibility into metrics and attendant carbon dioxide abatement.

Target users encompass approximately 500,000 smallholder farmers and 200 agro-industrial mills on the supply side. More than 20 biomass facilities require over 1 million tons of fuel annually on the demand side. Upwards of 50 local government units and Department of Energy offices engage for strategic planning. Anticipated impacts include an augmentation of waste-to-energy capacity from 182 megawatts to 282 megawatts. This generates ₱2 billion in supplementary farmer revenue, reduces 500,000 tons of carbon dioxide emissions, and achieves a 60% decline in open burning of residues within the first year (ASEAN Briefing, 2021; EnergyTracker Asia, 2024).

Conclusion

AgriPower PH reconfigures dispersed agricultural waste into a dependable clean energy feedstock. It simultaneously mitigates pollution, enhances rural livelihoods, and bolsters national energy resilience. Through digital orchestration of a fragmented supply chain the platform activates 150 megawatts of dormant capacity concentrated in pivotal provinces. This information technology intervention exemplifies how systematic innovation addresses intertwined waste management and energy imperatives within the Philippine context.

References

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